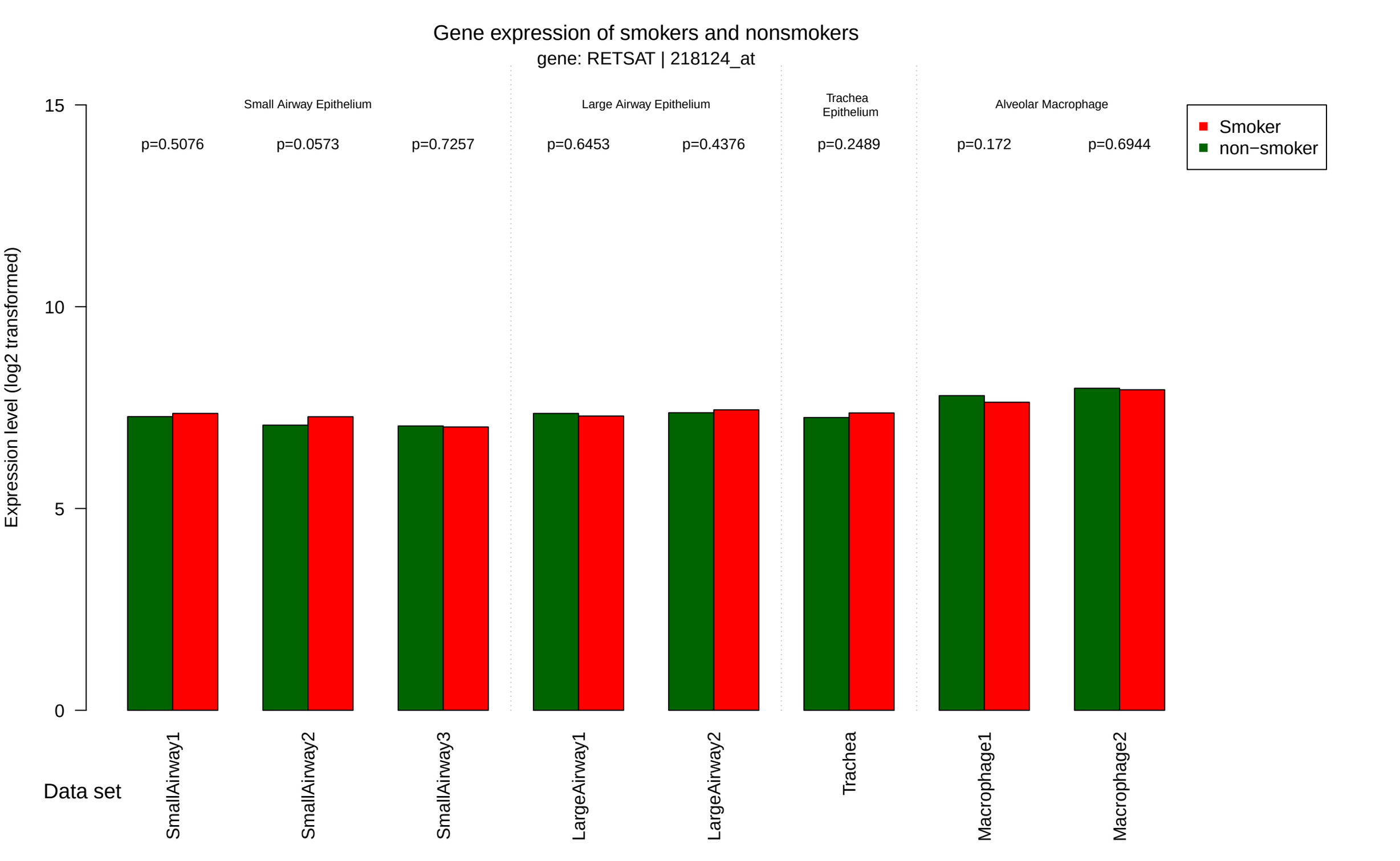
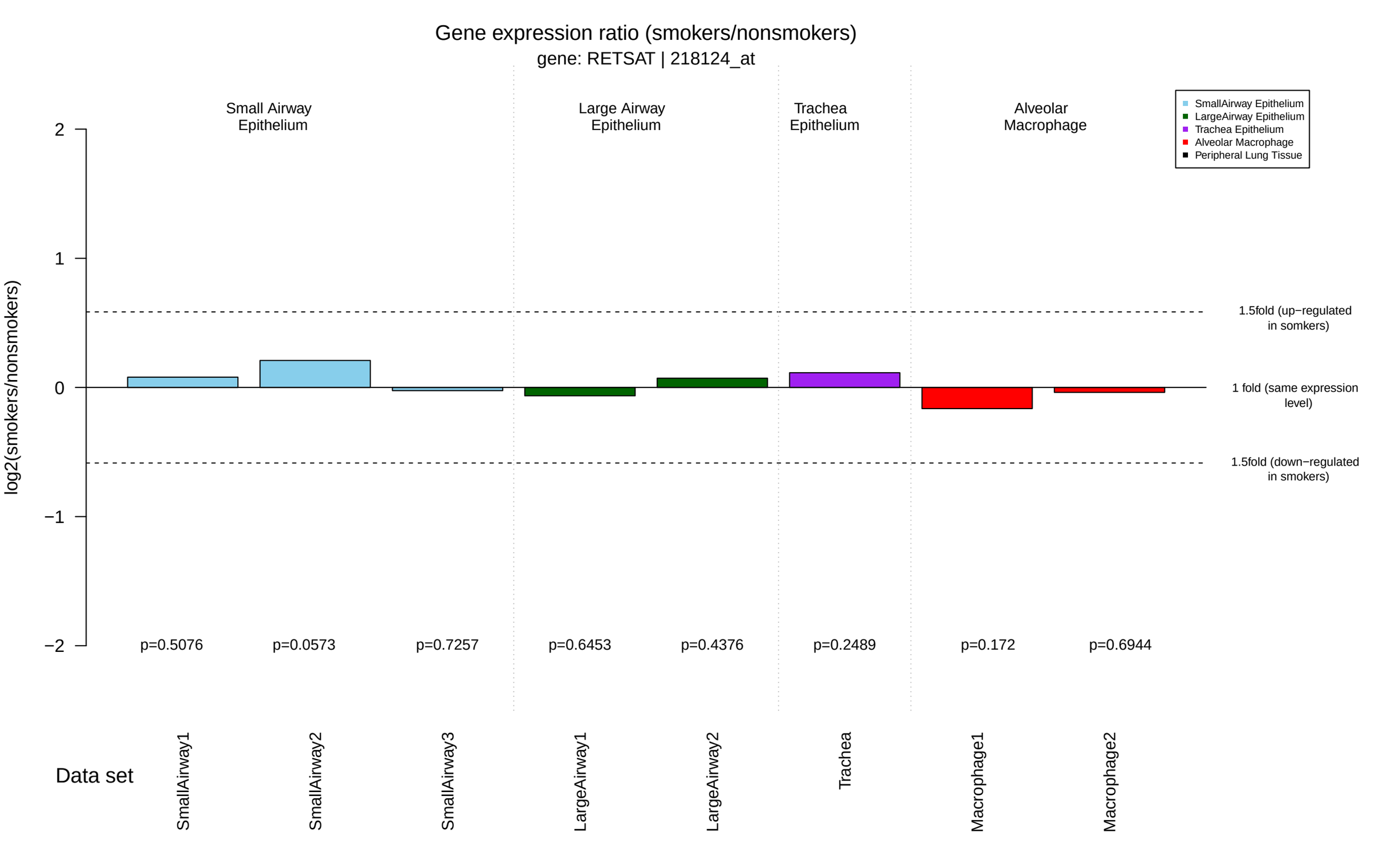
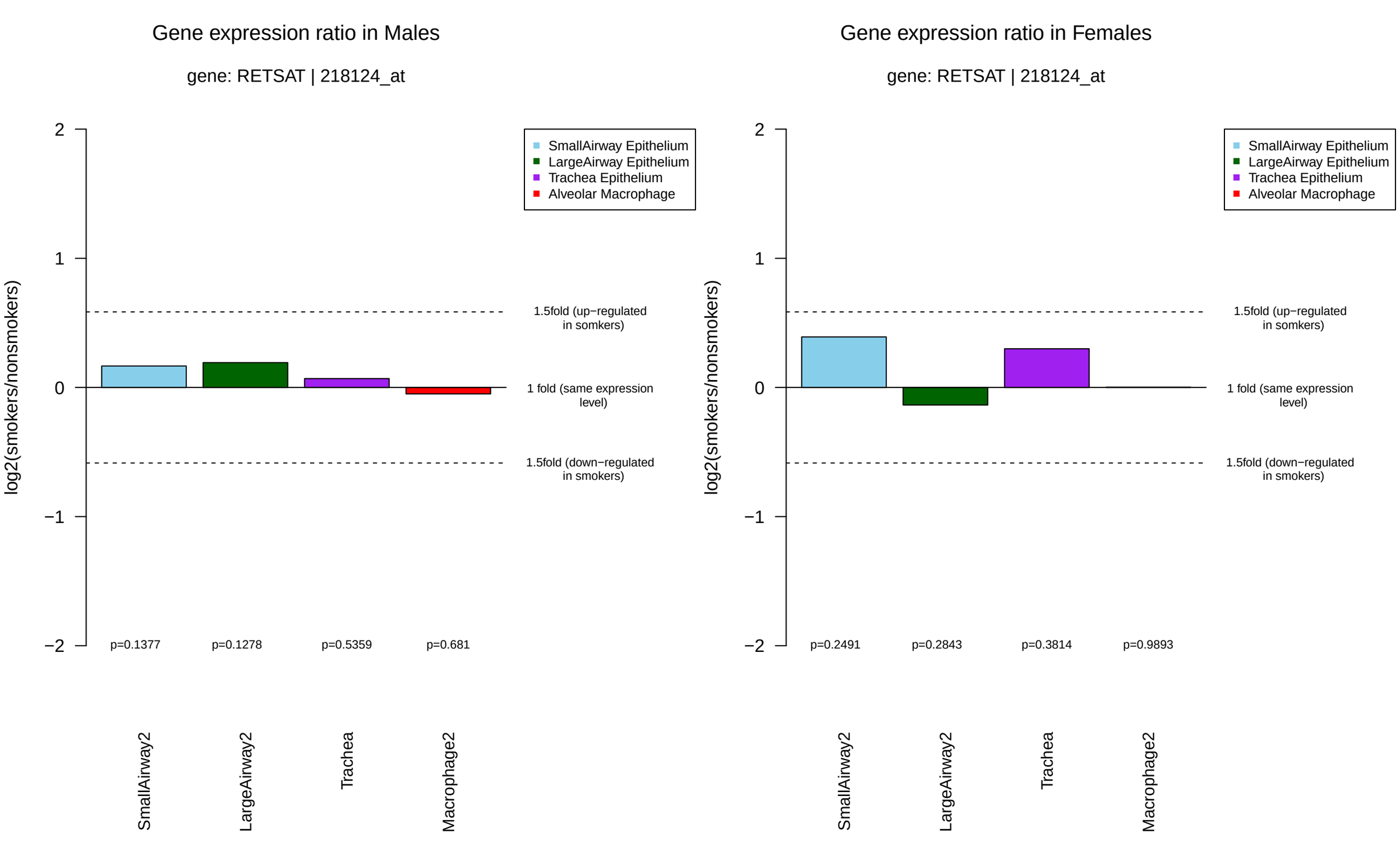
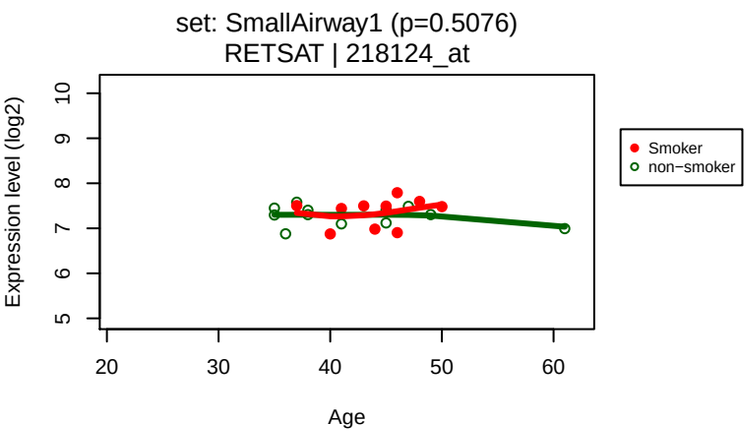
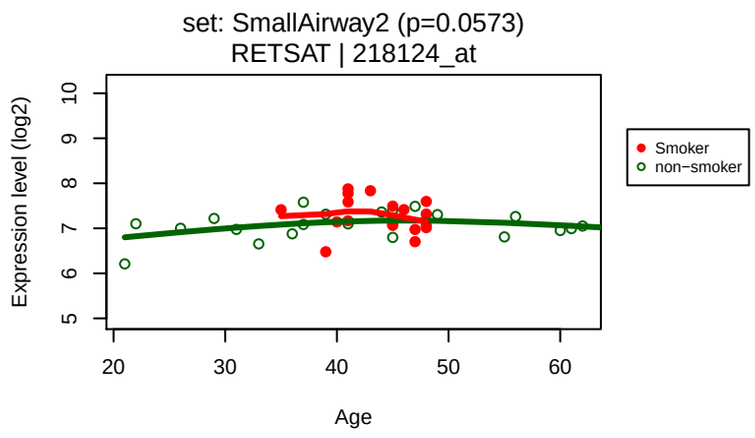
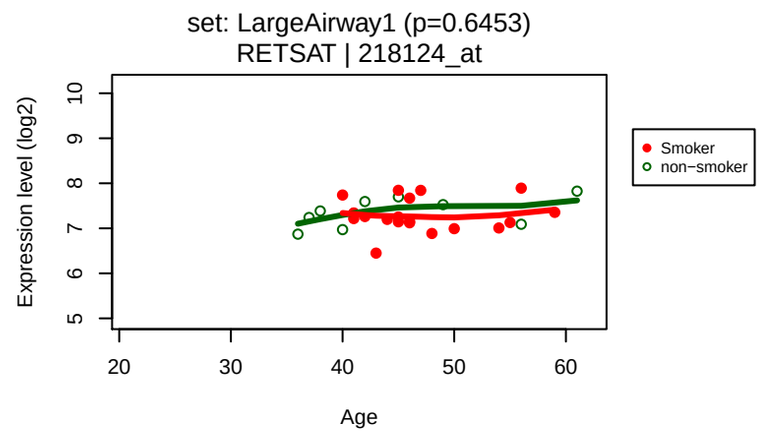

Gene expression of smokers and nonsmokers
gene: RETSAT | 218124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5076
p=0.0573
p=0.7257
p=0.6453
p=0.4376
p=0.2489
p=0.172
p=0.6944
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

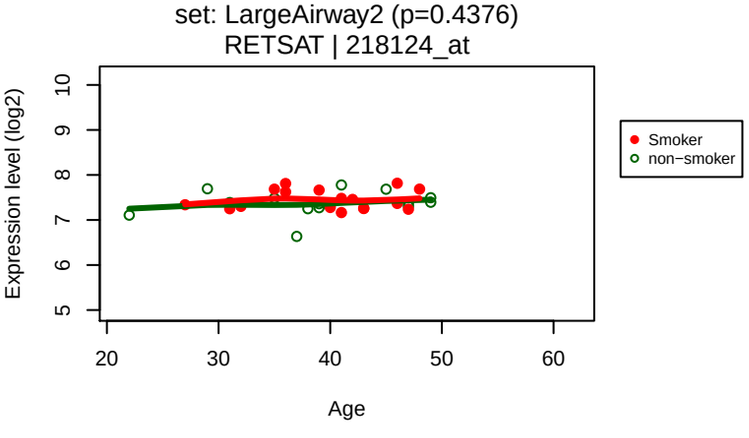

Gene expression ratio (smokers/nonsmokers)
gene: RETSAT | 218124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5076
p=0.0573
p=0.7257
p=0.6453
p=0.4376
p=0.2489
p=0.172
p=0.6944
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

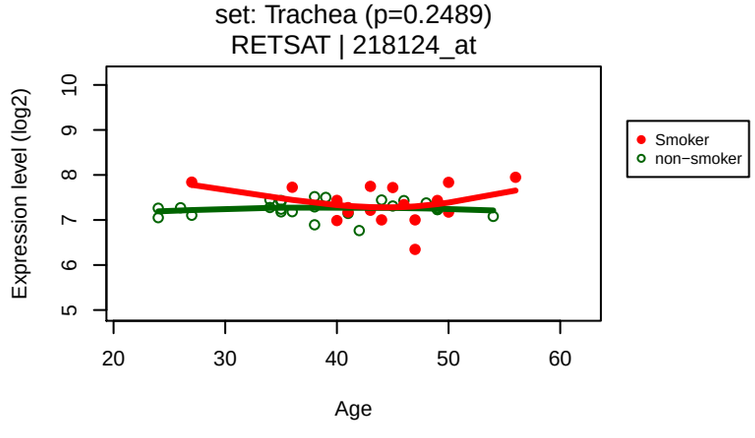

Gene expression ratio in Males
gene: RETSAT | 218124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1377
p=0.1278
p=0.5359
p=0.681
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RETSAT | 218124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2491
p=0.2843
p=0.3814
p=0.9893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

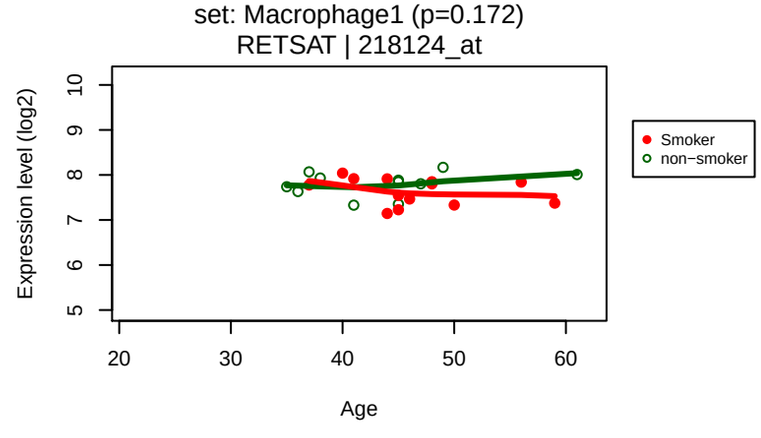

set: SmallAirway1 (p=0.5076)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker

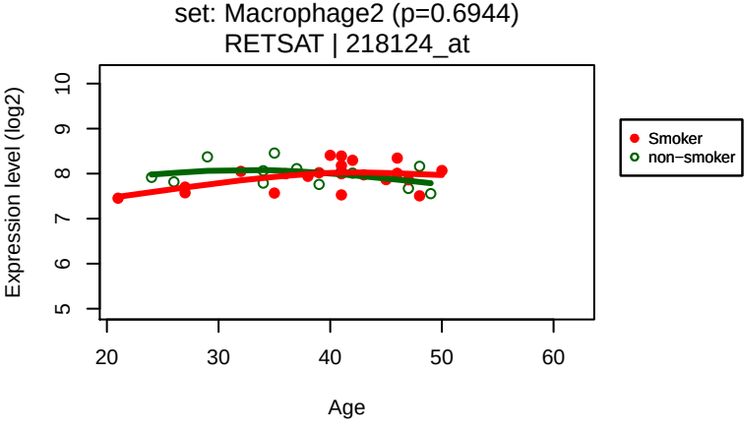

set: SmallAirway2 (p=0.0573)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker

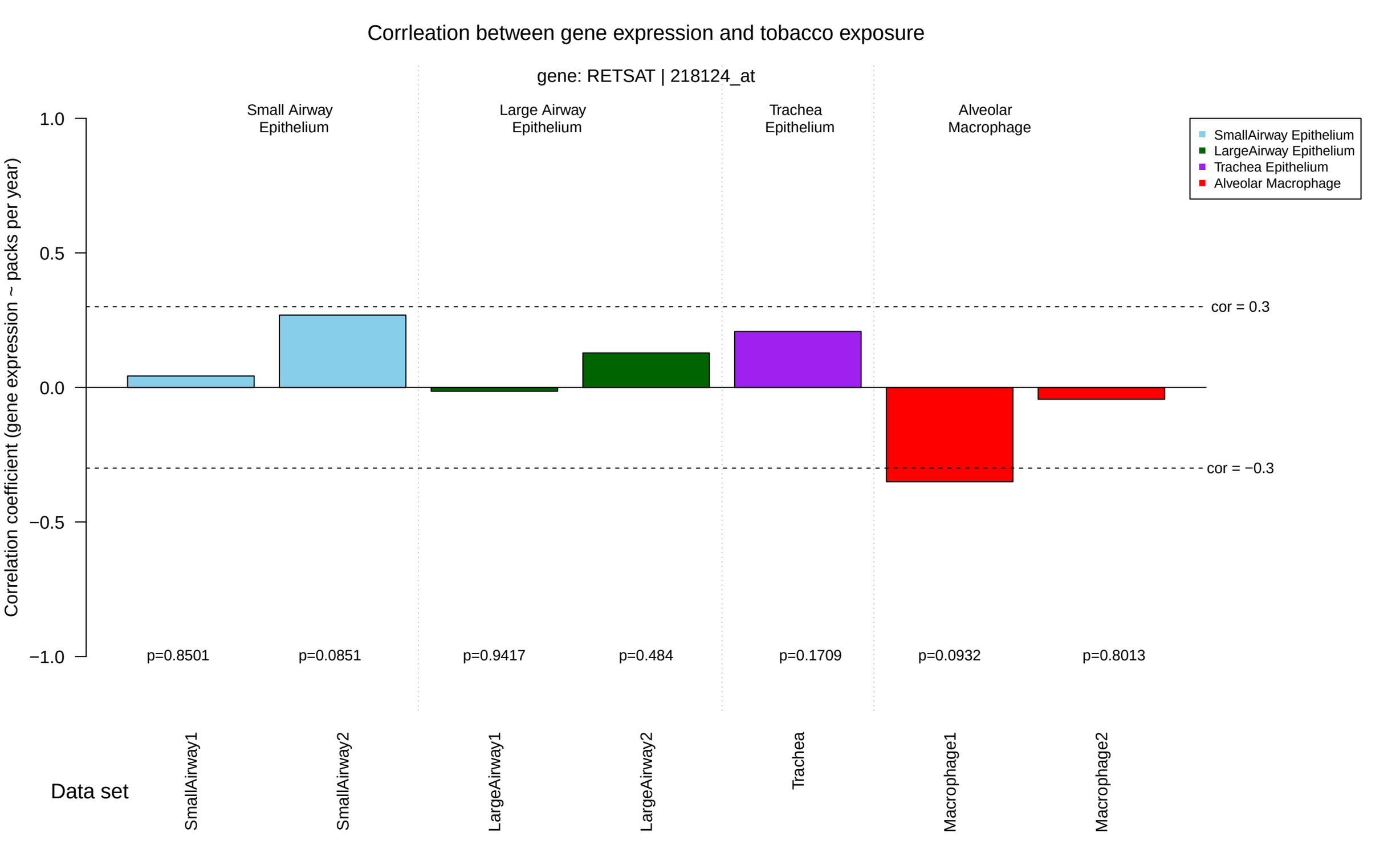

set: LargeAirway1 (p=0.6453)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4376)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2489)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.172)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6944)
RETSAT | 218124_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RETSAT | 218124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8501
p=0.0851
p=0.9417
p=0.484
p=0.1709
p=0.0932
p=0.8013
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2